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CS23621 Mobile Application Laboratory

SMART MEAL PLANNING

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BONAFIDE CERTIFICATE

Certified that this Project titled "**SMART MEAL PLANNING**" is the Bonafide work of "**AKASH N (230701019), ARYA SUBANANTH R K (230701039)**" who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

In an era of rising lifestyle diseases and increasing health consciousness, monitoring dietary intake has become essential for maintaining personal wellness. Nutrition Tracker is a comprehensive Android-based mobile application developed using Kotlin, Jetpack Compose, and Room Database, designed to empower users in managing their daily food consumption and nutritional goals. The application integrates multiple essential features including a meal tracking system, weekly meal planner, pantry manager, shopping list, and user profile with customizable fitness goals. It leverages a smart quantity parser using regex-based detection to automatically interpret food entries such as "100g chicken" or "2 dosa" and map them against a built-in nutrition data engine covering popular Indian and international foods.

The application follows the MVVM (Model-View-ViewModel) architecture pattern, ensuring a clean separation of concerns, testability, and scalability. Data persistence is achieved through a Room database with Flow-based real-time updates, while Coroutines handle all background operations efficiently. The user interface, built with Material 3 and Jetpack Compose, provides an intuitive, modern experience with dynamic color support, dark mode, circular progress indicators, and smooth animations. Nutrition Tracker stands as a practical, offline-first digital health companion that helps users make informed dietary decisions and achieve their fitness objectives — all within a single, accessible mobile platform.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL

Smart Meal Planning is an offline-first personal health and diet management application developed for Android using Kotlin, Jetpack Compose, and Room Database in Android Studio. Designed for health-conscious users and fitness enthusiasts seeking a streamlined approach to monitoring their dietary intake, the application integrates a suite of modules including a Home Dashboard, Meal Tracker, Weekly Planner, Pantry Manager, Shopping List, and User Profile. The system avoids cloud dependencies by utilizing a local Room database and SharedPreferences, ensuring fast performance and user data privacy.

Each module is engineered to function independently yet cohesively, supporting comprehensive nutrition management workflows. Users can log meals with detailed macronutrient breakdowns, receive daily progress indicators for calorie and macro goals, plan weekly meals, manage household pantry items, and maintain shopping lists — all within a single unified application. The interface is built with Material 3 composables, animated with smooth transitions, and optimized for usability across various screen sizes, while Jetpack Compose Navigation ensures a fluid user experience.

1.2 SCOPE OF THE WORK

This project focuses on the design and development of Smart Meal Planning, a lightweight and offline-capable Android application that centralizes essential dietary management tools into a single, intuitive platform. The scope of the work encompasses the implementation of multiple functional modules, including a Meal Tracker, Weekly Planner, Pantry Management system, Shopping List manager, and a User Profile with fitness goal configuration. Each

module is engineered using Kotlin in Android Studio and built on top of Jetpack Compose, Room Database, and MVVM architecture.

In addition, the project addresses UI/UX optimization through Material 3 design, dynamic colors, circular progress indicators, and responsive composable layouts. A Smart Quantity Parser using regex-based detection is implemented to automatically interpret user-entered food quantities (e.g., "100g chicken", "2 dosa") and map them to a built-in Nutrition Mapping Engine covering 12+ common foods including Indian staples. Special attention is given to database efficiency, real-time Flow-based updates, and Coroutine-backed background operations, making the application suitable for everyday mobile usage. Overall, Nutrition Tracker is designed to enhance the day-to-day dietary awareness of users without reliance on internet connectivity or external data storage.

1.3 AIM AND OBJECTIVES OF THE PROJECT

The primary aim of this project is to design and implement Smart Meal Planning, a comprehensive Android application that integrates multiple essential dietary management tools into a single, modular platform. This app is intended to support offline functionality and enhance user health awareness without relying on cloud services or internet connectivity. The core objective is to replace the fragmented experience of using separate applications for meal logging, calorie counting, grocery management, and fitness tracking, by offering a unified and efficient solution that operates entirely on the user's device.

To fulfill this aim, the project focuses on building individual modules for each key feature, including Meal Tracking, Weekly Meal Planning, Pantry Management, Shopping List, and User Profile with configurable fitness goals. Each module operates independently while maintaining seamless data integration — for instance, pantry stock levels can inform the shopping list, and tracked meals automatically update the dashboard's calorie and macro progress indicators. The application uses a Room database with DAO interfaces and a

Repository pattern to ensure all data is persisted reliably and efficiently on the device.

A user-centric design approach is employed, utilizing Jetpack Compose with Material 3 to deliver an intuitive and visually engaging experience across various screen sizes. The app prioritizes privacy by keeping all user data local, with no requirement for login via external services. Future scalability is also considered, with the modular MVVM code structure designed to support further enhancements, such as cloud backup, barcode food scanning, or AI-based meal recommendations.

The project also focuses on promoting healthier eating habits and improving overall lifestyle management through smart digital assistance. By providing real-time calorie tracking, macronutrient analysis, and progress visualization, the application helps users make informed dietary decisions every day. Reminder notifications for meals, water intake, and grocery restocking further enhance user engagement and consistency in maintaining fitness goals. The application is optimized for smooth performance on Android devices with minimal resource consumption, ensuring accessibility for a wide range of users. Proper input validation and error-handling mechanisms are included to improve reliability and data accuracy. In addition, the modular architecture of the application allows future integration of advanced technologies such as barcode scanning, cloud synchronization, wearable device connectivity, and AI-driven meal recommendations, making the system scalable, flexible, and future-ready.

CHAPTER 2

LITERATURE SURVEY

A mobile Smart Meal logging application using local SQLite storage and barcode scanning is introduced in [1]. It allows users to track calorie and macronutrient intake with a searchable food database. A user study demonstrates a 28% improvement in dietary adherence among regular users. Data persistence relies on on-device storage, ensuring offline accessibility. This study supports the Meal Tracking and Nutrition Mapping modules of Nutrition Tracker.

A smart food quantity recognition system using regex parsing and unit normalization on Android is proposed in [2]. It processes natural language food entries such as "250ml milk" or "1 cup rice" and resolves them to standardized nutritional values. Tests show 94% parsing accuracy across 500 sample entries. Parsed results are stored as structured records in SharedPreferences. This research directly supports the Smart Quantity Parser module in Nutrition Tracker.

An adaptive weekly meal planning system based on user calorie goals and historical meal patterns is presented in [3]. Using a rule-based scheduling engine and a visual calendar UI, it alerts users when meal slots are empty and tracks goal completion. AlarmManager is used for missing meal notifications, and plans are persisted using Room Database. Meal skipping decreased by 22% in longitudinal tests. This supports the Weekly Planner module in Nutrition Tracker.

A pantry management application using local JSON and expiry-based notification scheduling is detailed in [4]. Entries include item name, quantity, and expiry date, with a search-enabled interface for rapid retrieval. A user study showed significant reduction in food waste among participants. Efficient local

storage allows fast access and real-time inventory updates. This research guides the Pantry Management module in Nutrition Tracker.

A shopping list application with swipe-to-delete gestures, item completion tracking, and auto-suggestion capabilities is developed in [5]. It uses Room Database for persistent storage and Flow-based LiveData for real-time UI synchronization. A study showed a 35% reduction in forgotten grocery items. Suggestions are generated based on frequently purchased items history. This supports the Shopping List module in smart meal panning.

A fitness and health tracking application with real-time progress monitoring and personalized goal management is presented in [6]. The system allows users to monitor daily calorie intake, water consumption, and macronutrient balance through an interactive Android interface. It uses Room Database for offline data storage and Kotlin Coroutines for smooth background processing. Dynamic progress indicators and weekly analytics help users maintain healthy eating habits effectively. The application also provides reminders and notifications for meal tracking consistency. User feedback analysis showed improved engagement and better nutrition awareness among participants. Material 3 design components enhanced the visual experience and usability of the system. This study supports the Dashboard and Goal Tracking modules of Smart Meal Planning

CHAPTER 3

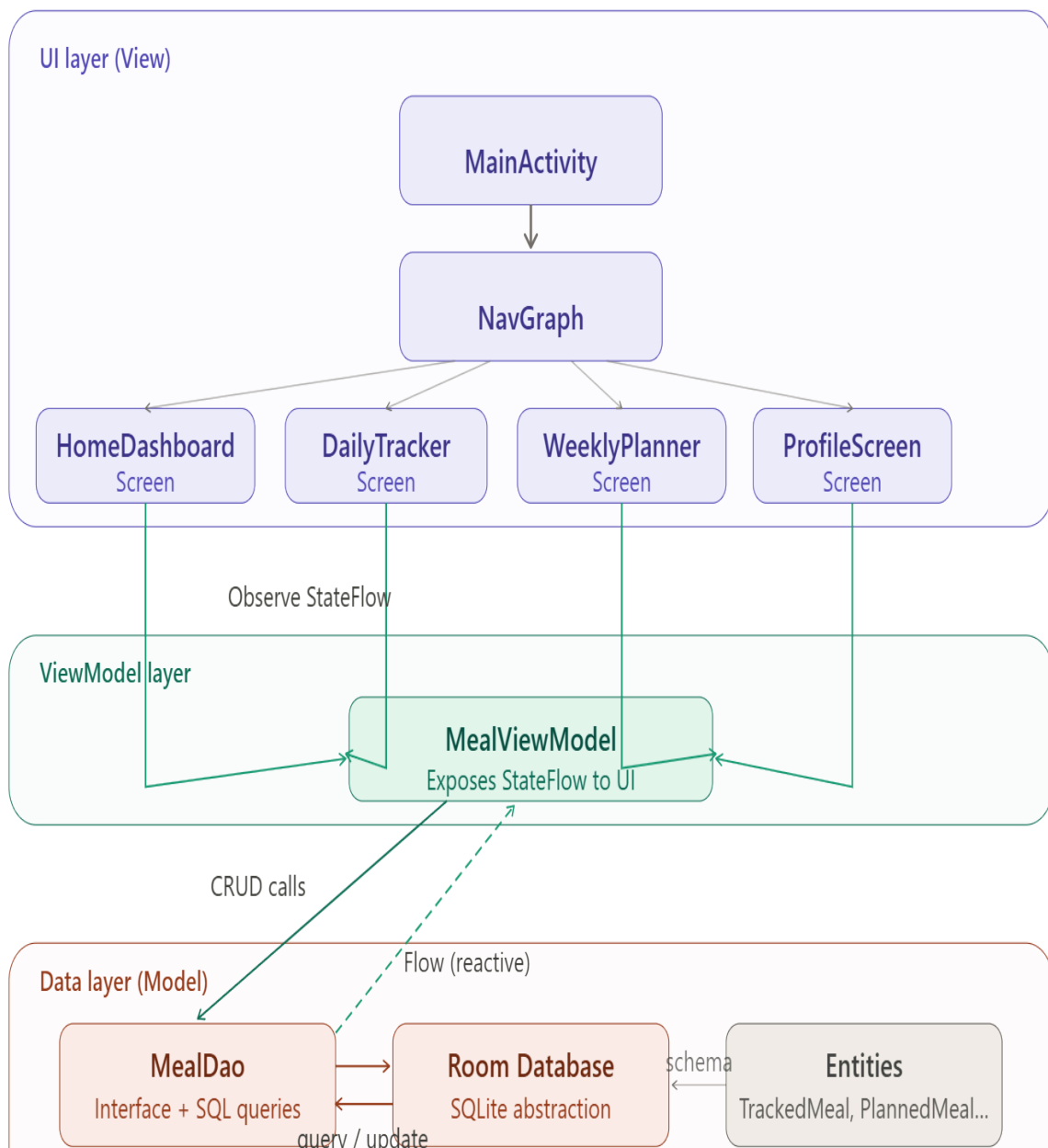
SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM.

The Smart Meal Planning App follows a layered MVVM architecture for efficient and offline-friendly performance. Each feature such as Home Dashboard, Meal Tracker, Weekly Planner, Pantry Manager, and User Profile has its own ViewModel and Compose screen.

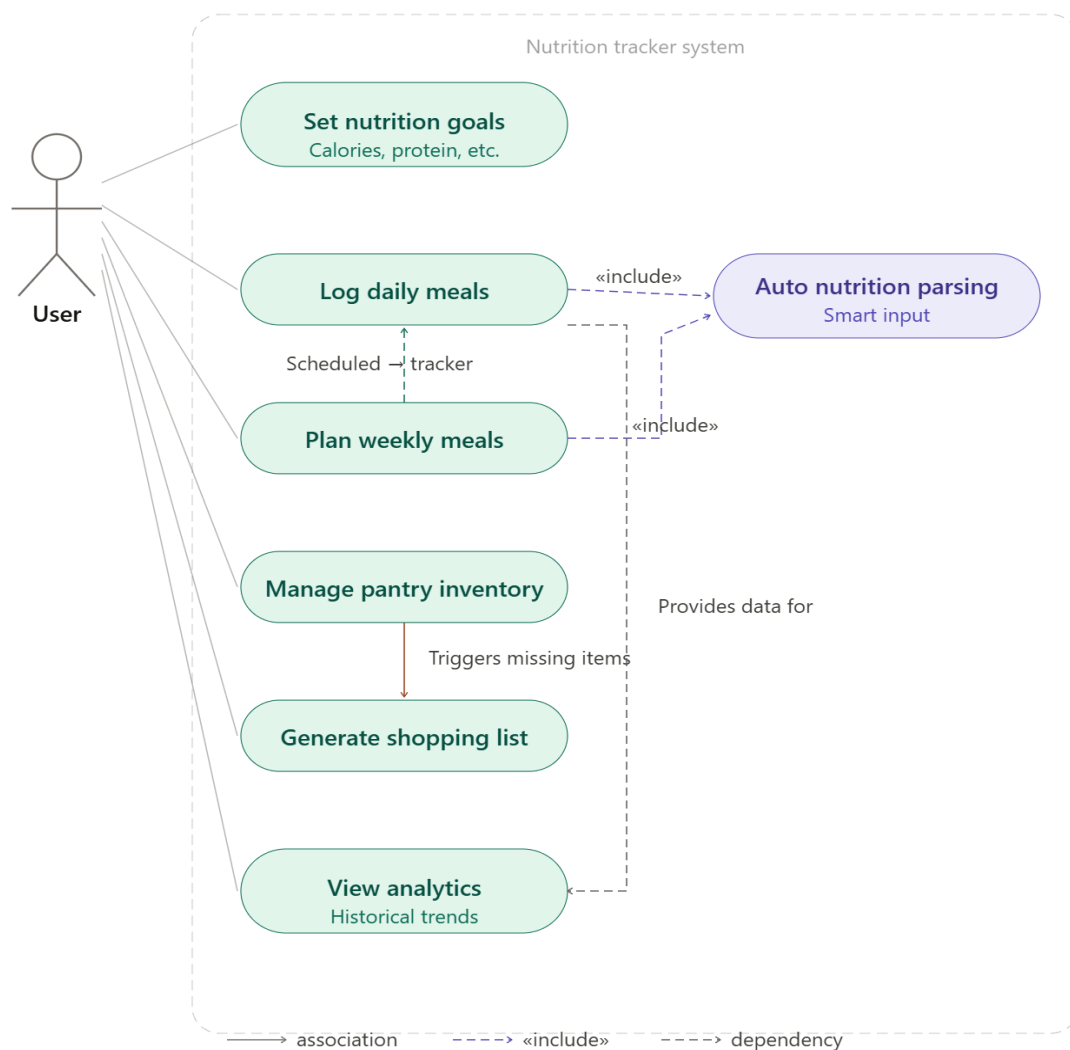
The Smart Meal application follows a modern MVVM architecture designed for better performance, scalability, and maintainability. The architecture is divided into three layers: the UI layer, ViewModel layer, and Data layer. Different colors are used in the diagram to clearly represent each layer. Features such as Home Dashboard, Meal Tracker, Weekly Planner, Pantry Manager, Shopping List, and User Profile are developed as separate composable screens connected with their own ViewModels. This structure helps in keeping the code organized and easy to manage.

The data flow in the architecture is represented using different arrow styles. Solid green arrows show direct communication between screens, ViewModels, and repositories, while dashed green arrows represent reactive data flow using Kotlin Flow from the database to the ViewModel. Faint gray arrows indicate navigation relationships handled through the centralized NavGraph. The use of Room Database, Coroutines, StateFlow, and Jetpack Compose ensures smooth UI updates, offline functionality, and responsive performance throughout the application.



3.2 USE CASE DIAGRAM

The Use Case Diagram for Smart Meal Planning outlines key interactions between the user and the system. Users can log meals via the Meal Tracker, view daily nutrition summaries on the Home Dashboard, and plan weekly meals using the Weekly Planner. The Pantry Manager allows users to add, search, and remove pantry items with expiry reminders, while the Shopping List supports adding, completing, and deleting grocery items. The User Profile enables configuration of calorie and macro targets for personalized fitness tracking. These functions work together seamlessly to offer a comprehensive, adaptive tool for managing dietary goals and daily nutrition.



3.3 HARDWARE SPECIFICATIONS

The Smart Meal Planning mobile application is designed to function smoothly on standard Android smartphones without requiring specialized hardware. The system's architecture ensures that core features — meal tracking, real-time macro progress calculation, weekly planning, and local database operations — can run efficiently on devices with moderate processing power. The following hardware specifications are recommended for optimal performance:

- Processor: Quad-core processor (e.g., Qualcomm Snapdragon 660 or higher, MediaTek Helio P60 or higher)
- RAM: Minimum 3 GB (Recommended: 4 GB for smoother animations and composable rendering)
- Storage: 32 GB internal storage or higher (to store Room database, user data, and app assets)
- Graphics: Integrated GPU with OpenGL ES 3.0 support (for Material 3 dynamic colors and animations)
- Operating System: Android 8.0 (Oreo / API Level 26) or higher
- Display: Full HD (1080 x 1920) or higher resolution with touchscreen support
- Other Requirements: No internet connection required for core functionality; optional for future cloud backup

These specifications ensure the app runs efficiently, allowing smooth meal logging, real-time progress updates, and seamless navigation while providing a high-quality user experience across a variety of Android devices.

3.4 SOFTWARE SPECIFICATIONS

The Smart Meal Planning mobile application is designed using a modular MVVM software architecture to ensure smooth operation and maintainability. The app integrates features for meal tracking, nutrition calculation, weekly planning, pantry management, and shopping. The software stack is composed of the following technologies:

- Frontend Framework: Kotlin + Jetpack Compose (for declarative UI and seamless experience across Android devices)
- UI Design: Material 3 with dynamic color theming and dark mode support
- Architecture Pattern: MVVM (Model-View-ViewModel) with Repository pattern
- Database: Room Database with DAOs, Flow-based queries, and Coroutine-backed transactions
- Asynchronous Operations: Kotlin Coroutines and StateFlow for background tasks and reactive UI updates
- Navigation: Jetpack Compose Navigation with a centralized NavGraph
- State Management: StateFlow and collectAsState() for real-time UI synchronization
- Build System: Gradle (Kotlin DSL) with dependency version management
- Development Environment: Android Studio Hedgehog or higher
- Target SDK: API Level 34 (Android 14); Minimum SDK: API Level 26 (Android 8.0)

CHAPTER 4

PROPOSED SYSTEM

The Proposed System for the Smart Meal Planning app is designed to integrate multiple health and dietary management tools into one seamless offline-first platform, providing users with efficient ways to monitor meals, track macronutrients, manage pantry inventory, and plan weekly nutrition. The system focuses on user-centric features such as real-time calorie tracking, smart food quantity parsing, weekly meal scheduling, and configurable fitness goal monitoring.

Key components of the proposed system include:

1. **Home Dashboard:** Displays daily calorie consumption and remaining budget, macro progress with circular indicators for protein, carbohydrates, and fats, water intake tracker, today's logged meals, and a weekly summary overview.
2. **Meal Tracker:** Allows users to log meals across four categories — Breakfast, Lunch, Dinner, and Snacks — with food name, quantity, and automatically calculated nutritional values using the built-in Nutrition Mapping Engine.
3. **Smart Quantity Parser:** Employs regex-based detection to interpret natural language food entries (e.g., "100g chicken", "2 dosa", "1 cup rice") and maps them to precise nutritional data.
4. **Weekly Planner:** Provides a calendar-style interface for scheduling meals across the week, with missing meal alerts and goal completion indicators per day.

5. **Pantry Manager:** Enables users to add, search, and remove pantry items with quantity tracking and expiry reminders to minimize food waste.
6. **Shopping List:** Provides a grocery list with add, complete, and swipe-to-delete functionality along with auto-suggestions based on pantry status.
7. **User Profile:** Allows users to set and update personal calorie and macro targets, fitness goals, and profile information, persisted locally via Room Database.

The system integrates these modules into a cohesive MVVM-based experience, ensuring all user data is stored securely and accessed efficiently through Room Database with Flow-based real-time updates. The user interface is built using Jetpack Compose with Material 3 theming, ensuring responsiveness, modern aesthetics, and a user-friendly experience across all supported Android devices.

CHAPTER 5

MODULE DESCRIPTION

The Smart Meal app is designed with several modules to improve health management through efficient meal logging, real-time macronutrient tracking, weekly planning, pantry management, and personalized goal setting. Below is an overview of each module:

1. Home Dashboard Module

The Home Dashboard serves as the central hub of the application, displaying a real-time summary of the user's daily nutritional intake. Features include a daily calorie tracker showing consumed versus target calories, circular progress indicators for protein, carbohydrates, and fat macros, a water intake tracker with increment/decrement controls, a list of today's logged meals grouped by type, and a weekly summary card showing day-by-day calorie achievement. Data is sourced in real-time from the Room database via StateFlow-backed ViewModels.

2. Meal Tracking Module

The Meal Tracker allows users to log food items under four meal categories — Breakfast, Lunch, Dinner, and Snacks. Each entry includes the food name, quantity input, and automatically calculated calories, protein, carbs, and fat values. The Smart Quantity Parser processes natural language inputs using regex patterns to detect quantity and unit (e.g., grams, pieces, cups, ml), which are then resolved against the built-in Nutrition Mapping Engine. Logged meals are persisted in the Room database and reflected instantly on the Dashboard.

3. Weekly Planner Module

The Weekly Planner provides a calendar-style layout for scheduling meals across all seven days of the week. Users can assign meal types and food items to specific days, view goal completion indicators per day, and receive alerts for days with missing or incomplete meal plans. Planned meals are stored as `PlannedMeal` entities in the Room database and retrieved via Flow-based DAO queries for real-time calendar rendering.

4. Pantry Management Module

The Pantry Manager enables users to maintain an inventory of household food items with quantity tracking and expiry date management. Users can add new pantry items with name, quantity, and expiry date, remove consumed items, search for specific items, and receive expiry reminders through local notifications. Pantry data is stored as `PantryItem` entities in Room, with search powered by DAO SQL LIKE queries.

5. Shopping List Module

The Shopping List module provides users with a simple and efficient way to manage grocery purchases. Items can be added with name and quantity, marked as purchased with a checkbox, and deleted using swipe gestures. Auto-suggestions are generated based on pantry items running low on stock. Shopping data is persisted as `ShoppingItem` entities in Room, with real-time UI updates driven by Flow collection in the ViewModel.

5.1 KEY BENEFITS OF PROPOSED SYSTEM

Comprehensive Nutritional Awareness:

The app offers an integrated, real-time view of daily calorie and macronutrient intake, helping users make informed dietary decisions and maintain nutritional balance throughout the day.

Smart Food Entry with Quantity Parsing:

With the regex-based Smart Quantity Parser and built-in Nutrition Mapping Engine covering Indian and international foods, users can log meals quickly using natural language inputs without manual nutrition lookup.

Unified Meal and Grocery Management:

The combination of the Weekly Planner, Pantry Manager, and Shopping List provides a complete food lifecycle management experience — from planning and stocking to purchasing and consumption tracking.

Offline-First with Real-Time Updates:

Powered by Room Database and Kotlin Flow, the app delivers instant UI updates on all data changes without any internet dependency, ensuring reliability and data privacy for all users.

Modern and User-Friendly Interface:

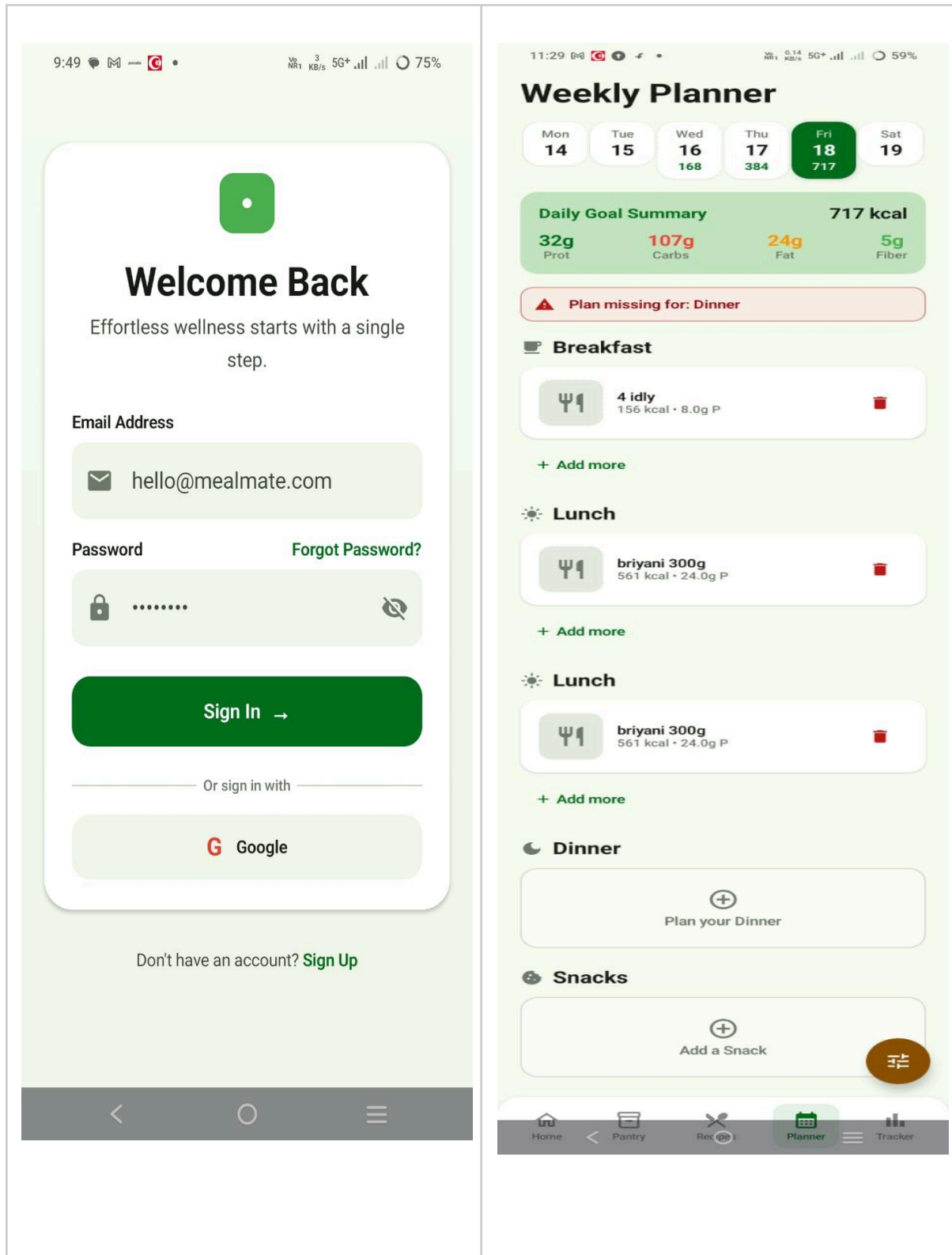
The application uses Jetpack Compose and Material 3 design components to provide a clean, responsive, and visually appealing user experience with smooth navigation and dynamic UI updates.

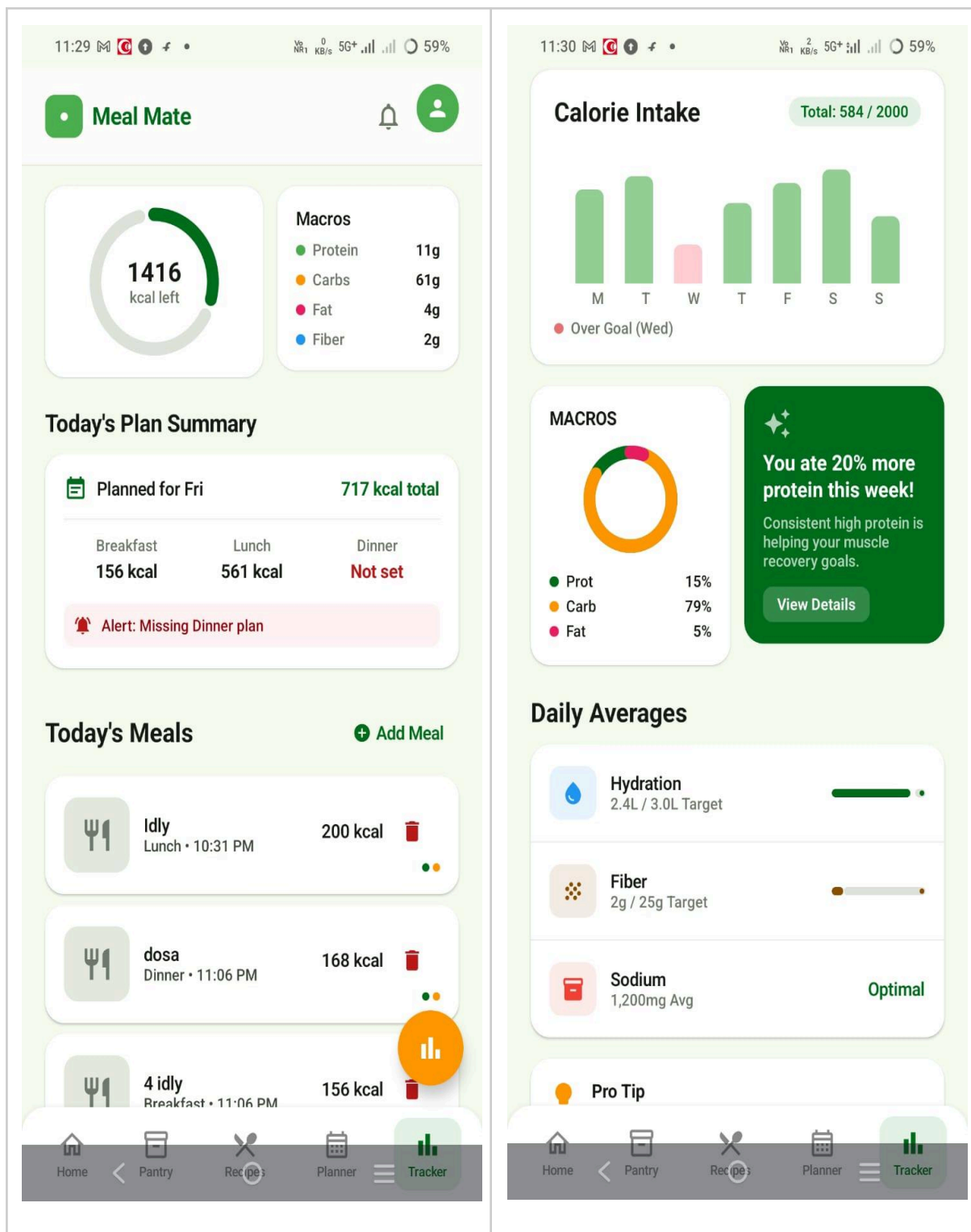
Scalable and Maintainable Architecture:

The MVVM-based modular architecture improves code organization, simplifies debugging, and allows future feature enhancements such as AI-based recommendations, cloud synchronization, and wearable device integration.

CHAPTER 6

OUTPUT





CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

The "Smart Meal Planning" application provides users with an all-in-one solution for managing daily dietary intake, setting nutritional goals, planning weekly meals, and organizing grocery shopping. With features like smart food quantity parsing, real-time macro tracking, a built-in nutrition database for Indian and international foods, and an offline-first Room database backend, the app aims to improve health awareness by integrating essential dietary management tools in a modern, user-friendly Jetpack Compose interface. The app's offline architecture ensures users can monitor their nutrition anytime and anywhere without relying on internet connectivity.

Future Enhancements:

1. **Barcode Food Scanner:** Integrating a camera-based barcode scanner to automatically fetch nutritional data for packaged food products from an open nutrition database.
2. **AI-Based Meal Recommendations:** Implementing AI-driven personalized meal suggestions based on the user's nutritional goals, dietary history, and macro deficits.
3. **Cloud Synchronization:** Adding optional Firebase-based backup and synchronization to allow users to securely access their nutrition data across multiple devices.
4. **Wearable Device Integration:** Connecting with fitness wearables such as Wear OS and Mi Band to automatically track calorie burn data and dynamically adjust daily nutritional targets in real time.
5. **Recipe Management Module:** Providing users with the ability to create, save, and manage custom multi-ingredient recipes with automatic calculation of total nutritional values for the complete dish.

CHAPTER 8

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